

SMART INDIA HACKATHON 2025



- **SIH-1605**
- **Women Safety Analytics** – Protecting Women from safety threats.
- **SafeHer:** AI-Powered Women Safety Analytics
- **PS Category- Software**
- **Team ID-**
- **Team Name - SAHAAYA**



Women Safety Analytics – Protecting Women from safety threats.



Background:

- Rising crimes against women demand real-time safety solutions.
- Need for AI-driven surveillance to detect threats early.

How it Works

- Monitors number of men & women in an area.
- Detects anomalies (lone woman at night, unusual gestures, crowd risk).
- Generates alerts to women so that she can act quickly.

Advantages

- Real-time monitoring & early detection.
- Prevents escalation of incidents.
- Identifies crime hotspots for better safety.
- Creates safer public environments.

Key Features

- Person detection & gender classification.
- Lone woman detection at night and crowd risk analysis (woman surrounded by men).
- Unusual gesture recognition and hotspot identification.
- Supports police intervention and builds trust in smart city safety systems.



Methodology (Step-wise Process)

1. Data Collection

Input: CCTV cameras, IoT sensors, microphones.(Capture real-time environment data).

2. AI-Powered Analysis

Person Detection ,Gender Classification → male/female ratio detection.

3. Decision Engine

Evaluates risk thresholds (e.g., lone woman at night + group of men nearby).Classifies as Normal / Suspicious / Threat.

4. Alert Generation

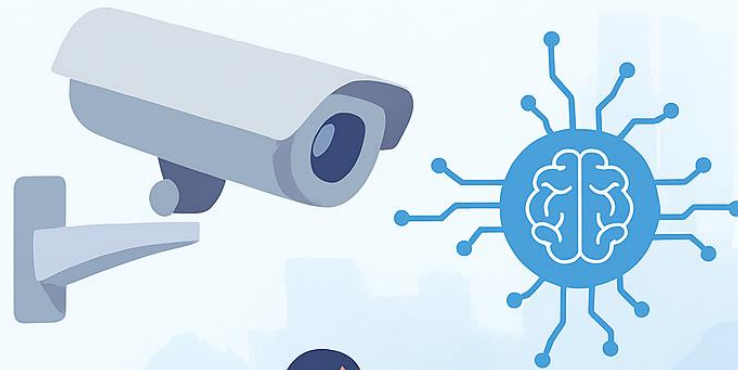
Automatic SOS alerts sent to:Women,Police stations,Registered emergency contacts.

5. Emergency Response

Law enforcement receives live feed+location.Triggers: alarms, lights, drone deployment (future scope).

FEASIBILITY AND VIABILITY

FEASIBILITY AND VIABILITY



- ✓ Technically feasible with existing AI/ML, CCTV, and IoT hardware
- ✓ Can integrate with already deployed surveillance systems

POTENTIAL CHALLENGES AND RISKS



- ⚠ False positives/negatives in detection (e.g., misclassification)
- 🔒 Privacy concerns in surveillance
- 📶 Infrastructure limitations (low-quality cameras, poor connectivity)
- 👤 User adoption and acceptance by authorities

STRATEGIES TO OVERCOME CHALLENGES

- ⚙ Continuous model training with diverse datasets to reduce errors
- ↑ Deploy edge devices to handle low-bandwidth scenarios

IMPACT AND BENEFITS



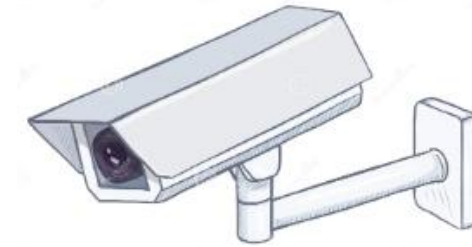
❖ Potential Impact:

For women in public spaces, workplaces, and institutions equipped with CCTV surveillance, this solution offers:

- **Enhanced Safety & Security** – Real-time monitoring and AI-driven threat detection ensure women feel protected while commuting, working late, or moving in high-risk zones.
- **Quick Emergency Response** – Automatic alerts to guardians reduce response time during potential threats.

❖ Benefits:

- **Crime Reduction:** Proactive threat detection deters harassment, assault, and unsafe activities in public spaces.
- **Improved Public Infrastructure:** Transforms CCTV from passive monitoring to active safety, increasing its social value.



- ❖ <https://drive.google.com/file/d/1JMiuf0SegdwO5elgKIMh9AC31U3eXZI7/view?usp=sharing>